

How performance pressure influences firms' cross-boundary growth: the moderating effect of managerial discretion

Firms' cross-boundary growth

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Abstract

Purpose – In today's complex and rapidly changing business environment, cross-boundary growth is increasingly critical to the survival or even success of organizations. The purpose of this study is to examine the forming mechanism of firm's cross-boundary growth by integrating the two important antecedent factors of performance pressure and managerial discretion into a united framework and theoretically analyze the direct role of performance pressure on firm's cross-boundary growth as well as reveal the moderating role of managerial discretion. Also, the authors select listed manufacturing companies in China as samples to empirically test the research hypotheses.

Design/methodology/approach – The authors design a multiple regression model to perform empirical analysis by using a panel of 4,002 year-observations in 1,334 listed manufacturing companies between 2013 and 2016. The sample data sources mainly come from the Wind Database, which is mainland China's leading financial database and software services provider. The hypotheses proposed are tested by adopting a panel data set of the listed manufacturing companies of China.

Findings – Empirical results show that performance pressure has a positive effect on the cross-industry growth and cross-domestic regional growth but a negative effect on the cross-international regional growth, and managerial discretion has a different moderating effect. Specifically, capital intensity strengthens the positive effect of performance pressure on cross-industry growth but weakens the negative effect of performance pressure on cross-international regional growth. State ownership enhances the positive effect of performance pressure on cross-domestic regional growth but decreases the negative effect of performance pressure on cross-international regional growth. CEO duality increases the negative impact of performance pressure on cross-international regional growth.

Practical implications – This study provides several implications for top executives, including how to dialectically consider the double-edged effect of performance pressure on cross-boundary growth of firms, create an appropriate environments of managerial discretion and design the types of cross-boundary growth strategies that top executives can follow in the volatility, uncertainty, complexity and ambiguity era.



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Originality/value – Although the relevant literature highlights the importance of performance pressure, it has not been related to the cross-boundary growth of firms. This paper makes an incremental contribution to the literature on the forming mechanisms of firm's cross-boundary growth by providing an important perspective of performance pressure to firm growth determinants and taking into account the moderating role of managerial discretion.

Keywords Performance pressure, Cross-boundary growth, Cross-industry growth, Cross-regional growth, Managerial discretion

Paper type Research paper

1. Introduction

To remain competitive and cope with uncertainty in the volatility, uncertainty, complexity and ambiguity (VUCA) era, more and more firms are seen voluntarily or under pressure moving into other areas with increasing frequency, and thus cross-boundary growth (hereinafter CBG) is increasingly critical to the survival or even success of firms. The concept of CBG first appeared in *Edsel Automobile*, an advertising magazine launched by Ford in 1957. Its original meaning is “transformation” and is now extended to “boundary spanning” and “crossover development” in new business models, market fields and locations (Birkinshaw *et al.*, 2017; Schotter *et al.*, 2017; Goerzen, 2018; Zott *et al.*, 2011). For example, the former “King of Shoes,” Guiren Bird, chose to cross-boundary engage in grain trading business due to successive plunges in performance. To cover the brand symbol to the majority of young people, Maotai cross-boundary layout ice cream market. On the contrary, Evergrande Group chose to abandon most of its marginal businesses when facing the debt crisis and narrow down cross-boundary areas. By observing the practice of firm development, it can be found that different realization states of the expected performance of firm may either induce or inhibit CBG, thus greatly influencing the choice of firm's cross-boundary behavior.

The existing literature has examined the effects of performance pressure on organizational behaviors from several aspects. For example, Jensen *et al.* (2019) demonstrated that performance pressure and ethical leadership interact to influence store-level incivility, and stores with higher incivility also had higher levels of shrink. Zhou *et al.* (2019) showed that pre-initial public offering (IPO) performance pressure has a negative impact on an IPO applicant's R&D investments. Spoelma (2021) noted that although performance pressure produces beneficial outcomes, it can also encourage cheating behavior, and personal values will have a positive intervention effect on them. Xu *et al.* (2021) stated that performance pressure produced both positive and negative effects on employees' in-role behaviors via approach motivation and avoidance motivation. Lv *et al.* (2022) analyzed the impact of inconsistencies among multiple performance pressure signals on the firm's R&D investment. Mai *et al.* (2022) investigated the exchange dynamics through which creative support provided by empowering leaders facilitates creative unethicity under conditions of high performance pressure. However, the existing researches mainly focus on the micro level such as enterprise R&D investment and employee organizational behavior, ignoring the impact of performance pressure on the enterprise strategy layer, especially without the connection with CBG.

According to organizational scholars, CBG refers to firms' business activities that move beyond organizational boundaries (Rosenkopf and Nerkar, 2001). These activities may involve growth activities across different business models, market fields and locations (Schotter *et al.*, 2017; Goerzen, 2018; Zott *et al.*, 2011). Based on the scope of growth, our study classifies two types of CBG: cross-industrial growth and cross-regional growth, where the latter can be further divided into cross-domestic and cross-international perspectives. Extant research has focused on understanding how organizational factors such as

resources, board compositions and networks and macro factors such as task environment, industrial competitiveness and institutional environment affect CBG (Hoskisson and Hitt, 1990; Hitt *et al.*, 2006; Li *et al.*, 2012; Du and Boateng, 2015). Some studies have also pointed out that early performance is one of the motivations for firms to choose CBG (Klueter and Monteiro, 2017), which only generally discuss whether and when the early performance affects subsequent cross-boundary activities, and rarely discuss what kind of CBG strategy should be adopted. Especially, the “double-edged sword” effect of performance pressure may have a completely opposite impact on a firm’s strategic choices, and we know less about what CBG decisions firms should make.

Furthermore, this study also provides a nuanced understanding of the associated contingencies by introducing and testing managerial discretion as critical moderators for the relationship between performance pressure and CBG. Hambrick and Finkelstein (1987) first introduced the concept of managerial discretion, as a way to understand whether, and when, executives have strategic leeway (Child, 1972). Managerial discretion refers to the latitude of options top managers have in making strategic choices (Hambrick and Finkelstein, 1987; Wangrow *et al.*, 2015). As originally conceptualized, a manager’s degree of discretion emanates from three levels: the environment, the organization and the individual (Hambrick and Finkelstein, 1987). This paper uses industry capital intensity, corporate ownership and CEO duality to measure each of the three levels of managerial discretion. Previous literature has shown that the degree to which executives influence decisions and their outcomes is moderated by different managerial discretion within the organization (Hambrick, 2007; Crossland and Hambrick, 2007; Baixauli-Soler *et al.*, 2021). When performance pressure appears, it is necessary to rely on the target of the firm, namely, the top managers, to perceive such pressure and then adopt appropriate response strategy according to their own experience and ability (Abhinav *et al.*, 2019). At this time, the managerial discretion is particularly important to the strategic choice.

To sum up, since performance pressure could produce paradoxical effect, we seek to explore why some firms handle the pressure better than others. Firms can benefit from knowing which situations are more inclined to respond to the pressure functionally. Therefore, the paper theoretically analyzes the influencing mechanism of performance pressure on firm’s CBG, as well as reveals the moderating mechanisms of managerial discretion. The subsequent section will review relevant literature and develop the hypotheses. This will be followed by the description of the methodology employed and the presentation of the empirical results. The paper will conclude with a discussion on the findings, implications and limitations of the study.

2. Theory and hypotheses development

2.1 Theoretical background

Performance pressure is a specific type of pressure based on the belief that striving for performance excellence is needed and that performance efforts will be scrutinized and tied to significant consequences (Mitchell *et al.*, 2018). The paper draws on the prospect theory and upper echelon view to analyze the influencing mechanism among performance pressure, managerial discretion and CBG of firms. First, the prospect theory, proposed by Daniel Kahneman, a professor of Princeton University and the winner of the Nobel Prize in Economics in 2002, described and predicted people’s behavior that are inconsistent with traditional expected value theory and expected utility theory in the decision-making process facing risks. Top managers choose and implement corporate strategies by evaluating the gap between actual performance and expected performance level, which refers to the reference point or minimum satisfaction value of decision-makers (Argote and Greve, 2007). Based on the results of performance feedback, the decision-maker tends to simply classify

the existing business results as success or failure and make strategic decisions accordingly (Cheng *et al.*, 2022). The theory emphasizes that senior executives' risk preferences will constantly change with the feedback results of performance during the process of making and implementing decisions. For example, Lim and Mccann (2014) showed that the worse a firm's actual performance was, the more likely it was to choose a risky but feasible strategy. Greve *et al.* (2010) pointed out that when firms are under high performance pressure, managers, motivated to avoid being fired, will give priority to illegal behaviors with low cost and rapid return to the desired performance level over R&D activities with uncertain output and long investment and return periods in the decision-making process. According to the related logic in the existing research, as a high-risk decision, the firm's CBG strategy will also be greatly affected by performance pressure. Firms need to consider their own specific context when they make the strategic decision to CBG, and performance pressure is one of the most important context factors. Therefore, this paper is based on prospect theory to analyze the mechanism between performance pressure and CBG strategy. Secondly, the upper echelon view, originally proposed by Hambrick and Mason (1984), emphasized that the senior managers' values and cognitive basis will have an important consequence on the strategic decision-making; the cognition and social psychology of senior managers can be inferred from the demographic characteristics of their members. Research showed that environmental and organizational factors are two important influences on managerial discretion; some organizations give their executives more freedom, and more changes and variety can be made in certain types of environments than in other types of environments (Hambrick and Finkelstein, 1987). Hambrick *et al.* (1993) showed that, compared with industries with low managerial discretion, the positive relationship between performance and firm commitment to the status quo is stronger in the industries with high managerial discretion. Rong *et al.* (2017) found a significant relationship between institutional ownership and international diversification in an empirical analysis of 197 large American companies. Li and Tang (2010) argued that managerial discretion positively moderated the relationship between CEO overconfidence and corporate risk-taking. In the case of expectation gap, the more shares a CEO holds, the more serious the loss of profits they face, which motivated them to quickly achieve high returns through illegal behavior to seek personal gain (Wang *et al.*, 2019). Therefore, with the help of upper echelon view, this paper explores whether managerial discretion plays a positive or negative role in the main effect research and explores the importance of managerial discretion to the long-term development and strategic choice of the firm.

2.2 Performance pressures and cross-boundary growth of firms

2.2.1 Performance pressure and firm cross-industry growth. Cross-industrial growth is a strategic behavior in which enterprises deviate from or even jump out of their current industries and enter into the other fields. When enterprises are in a disadvantageous position in the competitive pattern of the industry and it is difficult to surpass their rivals by adopting homogeneous competitive strategies, they will tend to obtain diversified competitive means through cross-industry innovation and challenge and subvert the traditional incumbent enterprises (Zhao, 2015). Ozcan and Eisenhardt (2009) used six small game companies in the USA as examples to describe how companies at the end of the industry can change existing competition rules through cross-industry development, thereby achieving catch-up. Ndofor *et al.* (2009) pointed out that, when firms are faced with profit pressure, they often adopt the cross-industry growth, including entering the upstream and downstream channels and obtaining the market and resources needed by firms. Li *et al.* (2021) pointed out that cross-industry growth in the crisis situation has a buffer effect on

protecting changes in enterprise value. In addition, firm cross-industry growth can avoid the fierce competition in the current market and, to a certain extent, increase the dependence on managers, which is convenient to increase compensation, obtain more power and reputation and reduce the potential risk of replacement (Rodenbach and Brettel, 2012). Cross-industry growth not only helps firm to relieve the performance pressure but also reduces the risk of top management being replaced. Therefore, we propose that:

H1. Performance pressure is positively related to cross-industry growth of firms.

2.2.2 Performance pressure and firm cross-regional growth. Cross-regional growth of firms refers to the expansion of business to other geographical regions, including cross-domestic and cross-international regions. Generally, there are significant differences in environmental conditions and potential risks between cross-domestic region growth and cross-international region growth of firms, so different regional expansion strategies will be implemented when firms were faced with performance pressure (Klueter and Monteiro, 2017). First, with the rise of performance pressure, the motivation of firms to carry out cross-domestic regions will increase. The main reasons are as follows: (1) performance pressure often arises from problems in business management, resource matching, product innovation or strategic choice. At this point, firms need to choose appropriate competitive strategies to cope with the challenges of customer preferences, product and market competition (Cheng *et al.*, 2022). Due to the imbalance of regional economic development in China, there are obvious differences in institutional environment and resource endowments within its different regions, so firms can match their life cycle stage with regional institutional environment conditions and choose the appropriate range of regional diversification (Zhang *et al.*, 2013). (2) when performance pressure becomes deteriorating, the social status and external legitimacy established in the past will be challenged or even threatened, and the firms often tend to be forced to re-examine the matching degree between their internal condition and external environment (Eggers and Suh, 2019), which may promote change or break through their current geographic region and carry out cross-domestic regional growth. Therefore, we propose that:

H2a. Performance pressure is positively related to cross-domestic regional growth of firms.

Second, compared with the cross-domestic-region growth of firms, cross-international region growth is beneficial for companies to acquire unique global knowledge (Almeida *et al.*, 2002; Santangelo and Phene, 2022), enabling them to innovate more effectively than their domestic competitors (Tushman and Scanlan, 1981), but it also hides higher risks and greater profit uncertainty and requires richer management experience and higher governance ability (Jiang and Holburn, 2018). To reduce uncertainty, manage bandwidth and reduce coordination costs, it is more likely for firms to reduce the unnecessary cross-international regional investment activities (Casson, 1997; Narula, 2014). In addition, to maintain their reputation and ensure business performance during their tenure, managers often adopt a growth strategy with high feasibility (Adner and Helfat, 2003). Since cross-international regional growth has higher requirements on capital, management experience and governance ability of firms, it should be more cautious to carry out cross-international regional growth when faced with performance pressure. Although negative performance feedback can prompt companies to become more open to cross-boundary, cross-international regional growth is usually an expensive process that requires large financial, technical, organizational and management resources (Castellani *et al.*, 2013; Levinthal and

March, 1981). That is, no matter from the perspective of motivation or ability, the possibility of firms implementing cross-international regional growth will be reduced with the increase of performance pressure. Therefore, we propose that:

H2b. Performance pressure is negatively related to cross-international regional growth of firms.

2.3 Moderating effect of managerial discretion

2.3.1 Industry capital intensity. Capital-intensive industries are characterized by large investment, more possession of capital, more technology and equipment, slow capital turnover and more difficult to timely adjust their current products or markets based on the current course of action and processes (Hambrick and Finkelstein, 1987). In some industry settings, managers enjoy a wide range of decision-making freedom, while in others, they may have little choice power (Dess and Beard, 1984; Li and Tang, 2010; Malhotra and Harrison, 2022). Highly capital-intensive industries pay more attention to maintain their existing capital, improve the efficiency of capital management and reduce the uncertain outcome of strategic changes (Rajagopalan and Finkelstein, 1992), which will constrain the manager's discretion choice and behavior space when making strategic decisions. Kim and Prescott (2005) pointed out that when faced with performance pressure, firms with low capital intensity have higher flexibility and greater discretion. Specifically, when the industrial capital intensity is high, it is more difficult for a firm to adjust its current strategy according to its own wishes. However, when the industrial capital intensity is low, firms hold more convertible resources and higher managerial discretion, which can provide more decision-making flexibility for the managers (Zacharias *et al.*, 2015) and make them more likely to make strategic decisions about firm's CBG. Therefore, we propose that:

H3a. Capital intensity moderates the relationship between performance pressure and firm's CBG. Specially, the higher level of capital intensity strengthens the positive relationship between performance pressures and firm cross-domestic growth but weakens the negative relationship between performance pressures and firm cross-international growth.

2.3.2 Corporate ownership. Corporate ownership refers to legal ownership of a business that determines who has the responsibility and obligation of controlling and operating the business (Ahmad and Omar, 2016). In emerging markets and Asian countries, there exists a mixture of different types of corporate ownership, including state-owned enterprises (SOEs), non-SOEs such as private enterprises and foreign enterprises (Garnaut *et al.*, 2005; Luo and Junkunc, 2008). Different types of corporate ownership have different access to resources and different approaches to business operation and decision-making (Baysinger *et al.*, 1991), which may affect the influence of performance pressure on firms' CBG. SOE is legal entity that undertakes commercial activities on behalf of government (Garnaut *et al.*, 2005). Since non-SOEs are either privately funded or funded by foreign capital, their major activity focuses on exploiting business opportunities to maximize profits (Ding *et al.*, 2007). Considering the nature of different types of ownership, we propose that the relationship between performance pressure and firms' CBG will be influenced by corporate ownership. The positive relationship between performance pressure and firms' CBG is likely to be weakened for firms with SOE ownership. First, SOEs have centralized decision-making style, and they also have privileged access to resources and market (Zhang *et al.*, 2017a; Du and Boateng, 2015). Empirical evidence shows that in a transition economy like China, SOEs

receive more compensation and support from the government under performance pressure than non-SOEs (Zhang *et al.*, 2013). Second, compared with no-SOEs, SOEs demonstrate a lower level of entrepreneurial orientation and risk-taking (Hunt, 2021). Given the high risk involved in CBG, and compared with non-SOEs, SOEs are less likely to pursue CBG. Therefore, we propose that:

- H3b.* Corporate ownership moderates the relationship between performance pressure and firm's CBG. Specifically, compared with state-owned enterprises, the performance pressure of non-state-owned enterprises has a greater effect on CBG of firms.

2.3.3 CEO duality. CEO duality means that the CEO serves as both chairman and president within one firm (Krause *et al.*, 2015). In modern corporate governance, whether CEO holds the two positions concurrently is an important institutional arrangement. Different dual roles of CEOs often lead to differential governance structures and result in varying degrees of managerial discretion (Tang, 2017). According to stewardship theory, the integration of two roles gives CEOs greater autonomy in making organizational change decisions, which can enhance their innovation risk-taking ability and management's approval rate for innovation decisions (Li and Tang, 2010). Yang *et al.* (2018) found that under a dual-role corporate governance, top managers have greater managerial discretion to exercise their will, thus reinforcing the effect of executive team on CBG. Furthermore, compared with the domestic regional environment, the international business environment is more complex and uncertain, so there are greater risks in cross-international regional growth (Jiang and Holburn, 2018; Xie *et al.*, 2019). For companies with dual CEO roles, shareholders have a greater impact on CEO's decisions, so they are more likely to prevent narcissistic CEO from taking a risky growth approach (Tang, 2017; Ali and Tauni, 2021). In this way, when the CEO holds both positions, firms will be more likely to implement the cross-domestic regional growth rather than cross-international regional growth out of the consideration for seeking benefits and avoiding disadvantages. Therefore, we propose that:

- H3c.* CEO duality positively moderates the relationship between performance pressure and firm's CBG. Specifically, CEO duality will strengthen the positive effect of performance pressure on firm cross-industry and cross-domestic regional growth, as well as strengthen the negative effect on firm cross-international regional growth.

Through literature review and the proposal of the above assumptions, this paper constructs a theoretical research model based on prospect theory and upper echelon view, as shown in Figure 1.

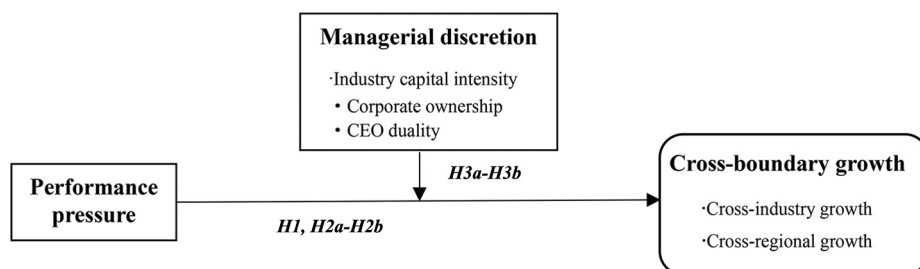


Figure 1.
Conceptual model

Source: Figure created by author

3. Research methodology

3.1 Sample and data sources

This study adopts the A-stock manufacturing listed companies on the main boards of Shenzhen and Shanghai in China from 2013 to 2016 as the sample. According to Jin (2015), after 2013, the Internet Era began to enter the stage of the combination of the virtual and real economy, which means increasing companies from the information and communications technology industries and the manufacturing industry started to seek the development by entering each other's areas. The sample data sources mainly come from the Wind Database, which is mainland China's leading financial database and software services provider and has built up a huge, complete and accurate data warehouse focused on the financial securities data, covering stocks, funds, bonds, foreign exchange, insurance, futures, financial derivatives, spot trade, macroeconomics, financial news and other fields. By the end of 2016, there were 2,173 listed companies in the manufacturing sector in the industry classification of China Securities Regulatory Commission. To better carry out data measurement and empirical analysis, samples are screened as follows: The companies of special treatment and SST are deleted to avoid the interference of outliers. Companies with missing data or incomplete disclosure are removed as well. Furthermore, since the firm's CBG is often the result of past performance pressure, the dependent variable data period (2014–2016) lags the other variables (2013–2015) by one year so as to reflect the lagged effect. Thus, the final sample is 1,334 firms with 3-year observation period from 2013 to 2016, and the number of firm-year observations is 4,002.

3.2 Model test

To test hypotheses H1, H2a and H2b, as well as test hypotheses H3a, H3b and H3c, the paper constructs the Formula (1) and Formula (2), respectively. The two empirical test models are as follows:

$$H_{i,t+1} = \alpha_0 + \alpha_1 EPG_{i,t} + \alpha_2 Cn_{i,t} + \varepsilon \tag{1}$$

$$H_{i,t+1} = \beta_0 + \beta_1 EPG_{i,t} + \beta_2 Mod_{i,t} + \beta_3 EPG_{i,t} * Mod_{i,t} + \beta_4 Cn_{i,t} + \varepsilon \tag{2}$$

Where $H_{i,t+1}$ denotes the dependent variable CBG; $EPG_{i,t}$ denotes the independent variable performance pressure, $Mod_{i,t}$ denotes the moderating variables of the three single-indicator, including industry capital intensity (CI_t), corporate ownership (OWN_t) and chair-CEO duality (CCD_t), respectively; $Cn_{i,t}$ denotes the control variables.

3.3 Variables measurement

3.3.1 Independent variable: cross-boundary growth

- Firm cross-regional growth

This paper adopts the Herfindahl Index developed by Berry (2014) to measure firm cross-regional growth. We draw on the classification methods of Rui *et al.* (2008) to divide the domestic geographical region of China into seven different sub-regions: East China, South China, North China, Central China, Northeast China, Northwest China and Southwest China. The formula is as follows:

$$H_g = \sum_1^n \left(\frac{S_i}{S} \right)^2 \quad (3)$$

Where S denotes the main business income of a firm, S_i denotes the main business income in the i th domestic region or country. H_g denotes the Herfindahl Index, and the value is distributed between 0 and 1. Generally, the larger the value, the less the distribution area, that is, the smaller the degree of firm cross-regional growth.

- Firm cross-industry growth

Similarly, the paper uses the Herfindahl index to measure firm cross-industry growth. According to the industry classification standard, if a firm sets foot into a new industry rather than requiring new entrants to dominate the industry, it means cross-industry growth (Duru and Reeb, 2002). The formula is as follows:

$$H_y = \sum_1^n \left(\frac{S_j}{S} \right)^2 \quad (4)$$

Where S denotes the main business income of a firm, S_j denotes the proportion of sales revenue of the j -th business in the total main business revenue. H_y denotes the Herfindahl Index, and the value is distributed between 0 and 1. Generally, the smaller the value, the more industries the firm is involved in and the higher the degree of cross-industry growth of the firm.

3.3.2 Dependent variable: performance pressure. Drawing on the existing methods (Chen, 2008; Lian *et al.*, 2015), the paper adopts the total return on assets to measure performance pressure. Specially, the performance pressure is measured by the difference between current and previous year's performance expectation. The calculation formula is as follows:

$$A_{i,t-1} = (1 - a_i)P_{i,t-2} + a_iA_{i,t-2} \quad (5)$$

Where a_i denotes the weight, which is distributed between 0 and 1. According to the previous result of robustness test, the a_i equals to 0.4. The performance pressure of firm i in $t - 1$ period is the difference between the actual performance $P_{i,t-1}$ and the expected performance $A_{i,t-1}$. The calculation formula is as follows:

$$EPG_{i,t} = P_{i,t-1} - A_{i,t-1} \quad (6)$$

The formula (6) indicates that with the increase of the $EPG_{i,t}$ values the actual performance $P_{i,t-1}$ continuously exceeds the expectation performance $A_{i,t-1}$. That is, there is an inverse relationship between $EPG_{i,t}$ and performance pressure.

3.3.3 Moderating variables. Drawing on the extant research, this paper measures managerial discretion from three dimensions including environmental, organizational and individual levels. Environmental discretion indicator is industry capital intensity (CI), measured by the ratio of the total assets to the main operating income of the sample companies at the end of the year. Organizational discretion indicator is corporate ownership (OWN), divided into two kinds of state-owned enterprises or non-state-owned enterprises. OWN is coded as 1 for state-owned enterprises and 0 for non-state-owned enterprises.

Individual discretion indicator is CEO duality (DUA), coded 1 if CEO simultaneously serves as a director and 0 for otherwise.

3.3.4 Control variables. This paper mainly selects the following three control variables: firm age (FAge), firm size (FSize) and the proportion of independent directors (IND). *Firm size (FSize)* is measured by the natural logarithm of total assets; *Firm Age (FAge)* is recorded as the difference between the sample observation year and the establishment time of the company. *The proportion of independent directors (IND)* is measured by the ratio of independent directors to board members at the end of the year for the sample company.

4. Empirical results analysis

4.1 Descriptive statistics and correlation coefficient matrix

Table 1 presents means, standard deviations and correlations among the variables. It shows that the performance pressure has a significantly positive effect on the cross-industry growth but a significantly negative effect on the cross-international regional growth. Meanwhile, managerial discretion plays a differentially moderating role in the relationship between performance pressure and CBG of firms. In addition, the correlation coefficient of the variables is less than 0.75, and most of them are less than 0.2, meaning that there is no obvious collinearity among the variables. The variance inflation factors for all variables are found to be less than 3.5, indicating that there is no significant multicollinearity among the variables. The above results preliminarily indicate that there exist significant relationship among performance pressure, managerial discretion and CBG of firms.

4.2 Main effect of performance pressure and firm’s cross-boundary growth

Table 2 provides the regressive results for hypotheses (H1), (H2a) and (H2b). As shown in Table 2, Model 1 and 2 reflect the regression results between performance pressure and firm’s CBG, and the regression coefficient is 0.032, significant at 0.05% level, indicating that performance pressure has a significantly positive effect on firm’s CBG, thus H1 is supported. Model 3 and 4 reflect the results between performance pressure and cross-domestic regional growth, and the regression coefficient is 0.023, significant at 0.05% level, indicating that performance pressure has a significantly positive effect on cross-domestic regional growth, thus H2a is supported. Model 5 and 6 reflect the results between performance pressure and cross-international regional growth, and the regression coefficient is -0.029, significant at 0.01% level, indicating that performance pressure has a significantly negative effect on cross-international regional growth, thus H2b is supported.

4.3 Moderating effect of managerial discretion

Table 3 provides the moderating results of capital intensity for hypothesis (H3a). As shown in Table 3, Model 1, 2 and 3 reflect the moderating effect of capital intensity on the relationship between performance pressure and cross-industry growth of firms. It shows that performance pressure has a significant positive effect on cross-industry growth ($\beta = 0.032, p < 0.05$), and their interaction also has a significant positive effect ($\beta = 0.004, p < 0.05$). Model 4, 5 and 6 reflect the moderating effect of capital intensity on the relationship between performance pressure and cross-domestic growth of firms. It shows that the interaction of capital intensity and performance pressure had a positive but insignificant effect on the cross-domestic regional growth ($\beta = 0.0001, p > 0.05$). Model 7, 8 and 9 reflect the moderating effect of capital intensity on the relationship between performance pressure and cross-international growth of firms. It shows that performance pressure has a significant negative effect on cross-international regional growth

Variables	Mean	SD	1	2	3	4	5	6	7	8	9
1. Cross-industry growth	0.650	0.374									
2. Cross-domestic region	0.504	0.294	-0.0425***								
3. Cross-international region	0.068	0.145	0.0397**	-0.3557***							
4. Performance pressure	-0.007	0.372	0.0279*	0.0260	-0.0769***						
5. Capital intensity	2.627	7.398	0.0200*	0.0442***	-0.0146**	0.2531***					
6. Ownership	0.360	0.480	-0.0213	0.0991***	-0.0806***	-0.0023	-0.0398**				
7. CEO duality	0.242	0.428	-0.0641***	-0.0155	0.0084*	0.0239	0.0090	-0.0224			
8. Firm age	2.653	0.388	-0.0930***	0.0988***	-0.0463***	-0.0031	0.0170	0.3221***	-0.0134		
9. Firm size	21.941	1.168	-0.0689***	0.0748***	-0.0595***	0.0437***	-0.0920***	0.3374***	0.0412***	0.1250***	
10. Independent director	0.378	0.101	0.0083	-0.0149	0.0153	-0.0219	0.0002	-0.0014	-0.0087	0.0030	0.0083

Notes: $N = 4,002$ observations; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, two-tailed

Source: Table created by authors

Table 1.
Descriptive statistics
and Pearson
correlations of the
study variables

Table 2.
Regression results of
performance
pressure on firm's
CBG

Variables	Cross-industry growth			Cross regional growth		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Constant	1.288*** (9.38)	1.303*** (9.48)	-0.165 (-0.46)	-0.043 (-0.12)	0.229*** (3.86)	0.204*** (3.43)
Firm size	-0.019*** (-3.19)	-0.020*** (-3.30)	-0.002 (-0.13)	-0.009 (-0.49)	-0.007** (-2.44)	-0.005* (-1.97)
Firm age	-0.087*** (-4.71)	-0.086*** (-4.69)	0.274*** (4.30)	0.284*** (4.44)	-0.011 (-1.33)	-0.011 (-1.39)
Firm's CBG	0.043 (0.76)	0.046 (0.82)	-0.017 (-0.35)	-0.015 (-0.30)	0.027 (1.44)	0.023 (1.28)
Performance pressure		0.032* (2.29)		0.023* (2.09)		-0.029*** (-6.70)
R ²	0.012	0.013	0.01	0.01	0.005	0.01
Adj-R ²	0.011	0.012	0.013	0.014	0.004	0.01
F value	1.98***	1.99***	2.21***	2.21***	4.41***	4.48***

Notes: $N = 4,002$ observations; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, two-tailed; t -values in parentheses
Source: Table created by authors

Variables	Cross-industry growth			Cross-domestic regional growth			Cross-regional growth			Cross-international regional growth		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 7	Model 8	Model 9
Constant	1.288*** (9.38)	1.303*** (9.48)	1.290*** (9.33)	-0.165 (-0.46)	-0.043 (-0.12)	-0.108 (-0.30)	0.229*** (3.86)	0.204*** (3.43)	0.219*** (3.68)	0.229*** (3.86)	0.204*** (3.43)	0.219*** (3.68)
Firm size	-0.019*** (-3.19)	-0.020*** (-3.30)	-0.020*** (-3.20)	-0.002 (-0.13)	-0.009 (-0.49)	-0.006 (-0.34)	-0.007** (-2.44)	-0.005* (-1.97)	-0.006* (-2.17)	-0.007** (-2.44)	-0.005* (-1.97)	-0.006* (-2.17)
Firm age	-0.087*** (-4.71)	-0.086*** (-4.69)	-0.087*** (-4.70)	0.274*** (4.30)	0.284*** (4.44)	0.283*** (4.44)	-0.011 (-1.33)	-0.011 (-1.39)	-0.011 (-1.37)	-0.011 (-1.33)	-0.011 (-1.39)	-0.011 (-1.37)
Firm's CBG	0.043 (0.76)	0.046 (0.82)	0.046 (0.82)	-0.017 (-0.35)	-0.015 (-0.30)	-0.017 (-0.35)	0.027 (1.44)	0.023 (1.28)	0.023 (1.24)	0.027 (1.44)	0.023 (1.28)	0.023 (1.24)
Performance pressure		0.032* (2.29)	0.034* (1.93)		0.023* (2.09)	-0.001 (-0.01)		-0.029*** (-6.70)	-0.038*** (-6.40)		-0.029*** (-6.70)	-0.038*** (-6.40)
Capital intensity			0.001 (1.10)			0.003*** (3.14)			-0.001*** (-3.86)			-0.001*** (-3.86)
Capital intensity *												
performance pressure			0.004* (1.42)			0.0001 (0.40)			0.002*** (3.63)			0.002*** (3.63)
R ²	0.012	0.013	0.013	0.01	0.01	0.015	0.010	0.010	0.012	0.010	0.010	0.012
Adjusted R ²	0.011	0.012	0.011	0.013	0.014	0.015	0.004	0.004	0.01	0.004	0.004	0.01
F value	1.98***	1.99***	1.99***	2.21***	2.21***	2.22***	4.41***	4.48***	4.53***	4.41***	4.48***	4.53***

Notes: N = 4,002 observations; *p < 0.05, **p < 0.01, ***p < 0.001, two-tailed; t-values in parentheses

Source: Table created by authors

Table 3.
Results of
moderating effect of
capital intensity

($\beta = -0.029, p < 0.001$), and their interaction has a significant positive effect ($\beta = 0.002, p < 0.001$). Thus, *H3a* is largely supported.

Table 4 provides the moderating results of ownership for hypothesis (*H3b*). As shown in Table 4, Model 1, 2 and 3 reflect the moderating effect of ownership on the relationship between performance pressure and cross-industry growth of firms. It shows that performance pressure has a significantly positive effect on cross-industry growth ($\beta = 0.032, p < 0.05$), but their interaction has no significant positive effect ($\beta = -0.066, p > 0.05$). Model 4, 5 and 6 reflect the moderating effect of ownership on the relationship between performance pressure and cross-domestic growth of firms. It shows that performance pressure has a significant positive effect on the cross-domestic regional growth ($\beta = 0.023, p < 0.05$), and their interaction has a significant negative effect ($\beta = 0.120, p < 0.05$). Model 7, 8 and 9 reflect the moderating effect of ownership on the relationship between performance pressure and cross-international growth of firms. It shows that performance pressure has a significant negative effect on the cross-international regional growth ($\beta = -0.029, p < 0.001$), and their interaction has a significant positive effect ($\beta = 0.033, p < 0.05$). Thus, *H3b* is largely supported.

Table 5 provides the moderating effect of CEO duality for hypothesis (*H3c*). As shown in Table 5, Model 1, 2 and 3 reflect the moderating effect of CEO duality on the relationship between performance pressure and cross-industry growth of firms. It shows that performance pressure has a significant positive effect on cross-industry growth ($\beta = 0.032, p < 0.05$), but their interaction has no significant effect ($\beta = -0.017, p > 0.05$). Model 4, 5 and 6 reflect the moderating effect of CEO duality on the relationship between performance pressure and cross-domestic growth of firms. It shows that performance pressure has a significant positive effect on the cross-domestic regional growth ($\beta = 0.023, p < 0.05$), but their interaction has an insignificant effect ($\beta = -0.020, p > 0.05$). Model 7, 8 and 9 reflect the moderating effect of CEO duality on the relationship between performance pressure and cross-international growth of firms. It shows that performance pressure has a significant negative effect on the cross-international regional growth ($\beta = -0.029, p < 0.001$), and their interaction has a significant positive effect ($\beta = 0.040, p < 0.001$). Thus, *H3c* is largely supported.

4.4 Robustness check

This paper further delays the time window of the effect of performance pressure on firm's CBG; that is, the dependent variables CBG lags other variables by two years. Table 6 presents the robustness check results of the performance pressure on firm's CBG. According to the results in the table, the regression coefficient β of performance pressure on the cross-industry growth of firms is 0.037 and $p < 0.05$. Thus *H1* is supported. The regression coefficient β of performance pressure on the cross-domestic regions growth is 0.033 and $p < 0.001$. Thus, *H2a* is supported. The regression coefficient β of performance pressure on the cross-regional growth is -0.030 and $p < 0.001$. Thus, *H2b* is supported.

Table 7 presents the robustness check of moderating results of managerial discretion. According to the results of Model 2 and Model 3, capital intensity still plays a significant positive moderating effect on the relationship between performance pressure and cross-industry growth of firms ($\beta = 0.023, p < 0.10$), while according to the results of Model 2, Model 4 and Model 5, the moderating effect of corporate ownership and CEO duality is not significant. Similarly, according to the results of Model 7 and Model 9, corporate ownership has a significant positive moderating effect on the relationship between performance pressure and cross-domestic regions growth ($\beta = 0.140, p < 0.10$), while capital intensity and CEO duality have no significant moderating effect. According to the results of Model 12, Model 13 and Model 15, capital intensity ($\beta = 0.011, p < 0.05$) and CEO duality ($\beta = 0.012, p < 0.01$) have a significant positive moderating effect on the relationship between

Variables	Cross-industry growth			Cross-domestic regional growth			Cross regional growth			Cross-international regional growth		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
Constant	1.288*** (9.38)	1.303*** (9.48)	1.388*** (9.47)	-0.165 (-0.46)	-0.043 (-0.12)	-0.035 (-0.10)	0.229*** (3.86)	0.204*** (3.43)	0.208*** (3.49)			
Firm size	-0.019*** (-3.19)	-0.020*** (-3.30)	-0.023*** (-3.66)	-0.002 (-0.13)	-0.009 (-0.49)	-0.010 (-0.56)	-0.007* (-2.44)	-0.005* (-1.97)	-0.006* (-2.15)			
Firm age	-0.087*** (-4.71)	-0.086*** (-4.69)	-0.095*** (-4.97)	0.274*** (4.30)	0.284*** (4.44)	0.288*** (4.50)	-0.011 (-1.33)	-0.011 (-1.39)	-0.010 (-1.21)			
Firm's CBG	0.043 (0.76)	0.046 (0.82)	0.045 (0.81)	-0.017 (-0.35)	-0.015 (-0.30)	-0.019 (-0.38)	0.027 (1.44)	0.023 (1.28)	0.025 (1.34)			
Performance pressure		0.032* (2.29)	0.034* (2.40)		0.023* (2.09)	0.026* (2.33)		-0.029*** (-6.70)	-0.031*** (-6.98)			
Corporate ownership			0.026 (1.62)			0.023 (0.42)			0.003** (0.65)			
Corporate ownership *												
performance pressure			-0.066 (-0.71)			0.120* (2.14)			0.033* (1.97)			
R^2	0.012	0.013	0.016	0.01	0.01	0.01	0.010	0.010	0.012			
Adjusted R^2	0.011	0.012	0.013	0.013	0.014	0.017	0.004	0.01	0.013			
F value	1.98***	1.99***	1.98***	2.21***	2.21***	2.20***	4.41***	4.48***	4.53***			

Notes: $N = 4,002$ observations; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, two-tailed; t -values in parentheses

Source: Table created by authors

Table 4.
Results of
moderating effects of
corporate ownership

Table 5.
Results of
moderating effect of
CEO duality

Variables	Cross-industry growth			Cross-domestic regional growth			Cross regional growth			Cross-international regional growth		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 8	Model 9	Model 9
Constant	1.288*** (9.38)	1.303*** (9.48)	1.297*** (9.43)	-0.165 (-0.46)	-0.043 (-0.12)	-0.081 (-0.22)	0.229*** (3.86)	0.204*** (3.43)	0.203*** (3.41)			
Firm size	-0.019*** (-3.19)	-0.020*** (-3.30)	-0.019*** (-3.14)	-0.002 (-0.13)	-0.009 (-0.49)	-0.008 (-0.43)	-0.007* (-2.44)	-0.005* (-1.97)	-0.005 (-1.95)			
Firm age	-0.087*** (-4.71)	-0.086*** (-4.69)	-0.087*** (-4.70)	0.274*** (4.30)	0.284*** (4.44)	0.290*** (4.53)	-0.011 (-1.33)	-0.011 (-1.39)	-0.012 (-1.44)			
Firm's CBG	0.043 (0.76)	0.046 (0.82)	0.042 (0.76)	-0.017 (-0.35)	-0.015 (-0.30)	-0.016 (-0.32)	0.027(1.44)	0.023 (1.28)	0.023 (1.25)			
Performance pressure		0.032* (2.29)	0.039* (2.14)		0.023* (2.09)	0.030 (1.76)		-0.029*** (-6.70)	-0.043*** (-6.99)			
CEO duality			-0.059*** (-4.58)			-0.015 (-1.31)			0.005 (1.12)			
CEO duality *												
performance pressure	0.012	0.013	-0.017 (-0.47)	0.01	0.01	-0.020 (-0.53)	0.010	0.010	0.040*** (3.19)			
R ²	0.011	0.012	0.016	0.013	0.014	0.009	0.010	0.010	0.012			
Adjusted R ²	1.98***	1.99***	2.00***	2.21***	2.21***	2.21***	4.41***	4.48***	4.49***			
F value												

Notes: N= 4,002 observations; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, two-tailed; t -values in parentheses
Source: Table created by authors

Variables	Cross-industry growth			Cross regional growth		
	Model 1	Model 2	Cross-domestic regional growth Model 3	Cross-regional regional growth Model 4	Cross-international regional growth Model 5	Model 6
Constant	0.450** (2.37)	0.444** (2.34)	0.051 (0.39)	0.028 (0.22)	0.273*** (3.91)	0.320*** (3.43)
Firm size	0.147*** (3.46)	0.168*** (3.91)	0.013* (2.27)	0.017** (2.96)	-0.007* (-2.06)	-0.009** (-2.83)
Firm age	0.006*** (0.27)	0.007*** (0.28)	0.071*** (3.98)	0.068*** (3.82)	-0.026 (-2.78)	-0.024*** (-2.63)
Firm's CBG	0.136 (0.99)	0.141 (1.03)	-0.019 (-0.53)	-0.019 (-0.53)	0.007 (0.31)	0.007 (0.33)
Performance pressure		0.037* (2.54)		0.033*** (8.01)		-0.030*** (-11.80)
R^2	0.003	0.003	0.016	0.017	0.011	0.019
Adjusted R^2	0.001	0.002	0.015	0.016	0.010	0.018
F value	2.41***	2.43***	19.20***	20.11***	11.74***	12.85***

Notes: $N = 4,002$ observations; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, two-tailed; t -values in parentheses
Source: Table created by authors

Table 6.
Robustness check:
regressive results of
performance
pressure and firm's
CBG

Table 7.
Robustness check:
moderating results of
managerial
discretion

Variables	Cross-industry growth				Cross-regional growth			
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Constant	0.450** (2.37)	0.444** (2.34)	0.489* (2.54)	0.417* (2.08)	0.429* (2.26)	0.051 (0.39)	0.028 (0.22)	0.034 (0.14)
Firm size	0.147*** (3.46)	0.168*** (3.91)	0.160*** (3.68)	0.175*** (4.00)	0.167*** (3.86)	0.013* (2.27)	0.017** (2.96)	0.015 (1.24)
Firm age	0.006*** (0.27)	0.007*** (0.28)	0.009*** (0.37)	0.010 (0.38)	0.006 (0.23)	0.071*** (3.98)	0.068*** (3.82)	0.064 (1.49)
Firm's CBG	0.136 (0.99)	0.141 (1.03)	0.142 (1.04)	0.144 (1.05)	0.152 (1.11)	−0.019 (−0.53)	−0.019 (−0.53)	−0.028 (−0.71)
performance pressure			0.039* (1.93)	0.036* (2.50)	0.037* (2.53)		0.033*** (8.01)	0.029 (1.44)
Capital intensity		0.037* (2.54)	−0.006 (−1.38)					−0.001 (−1.12)
Corporate ownership				−0.103 (−0.80)				
CEO duality			0.023* (2.08)		0.034 (1.34)			0.009 (1.01)
Capital intensity * performance pressure				0.214 (1.00)				
Corporate ownership * performance pressure					−0.051 (−0.17)			
CEO duality * performance pressure						0.016	0.017	0.018
R ²	0.003	0.003	0.005	0.004	0.004	0.015	0.016	0.016
Adjusted R ²	0.001	0.002	0.002	0.001	0.002	0.015	0.016	0.016
F value	2.41***	2.43***	2.43***	2.43***	2.42***	19.20***	20.11***	20.08***

Notes: N=4,002 observations; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, two-tailed; t -values in parentheses

Source: Table created by authors

(continued)

Variables	Cross-domestic regional growth			Cross regional growth			Cross-international regional growth		
	Model 9	Model 10	Model 11	Model 12	Model 13	Model 14	Model 15		
Constant	0.031 (0.12)	0.028 (0.21)	0.273*** (3.91)	0.320*** (3.43)	0.278*** (3.95)	0.318*** (4.57)	0.320 (4.63)		
Firm size	0.016 (1.28)	0.017** (2.95)	-0.007* (-2.06)	-0.009** (-2.83)	-0.007* (-2.21)	-0.009** (-2.97)	-0.009** (-2.82)		
Firm age	0.056*** (1.24)	0.067*** (3.81)	-0.026 (-2.78)	-0.024** (-2.63)	-0.025** (-2.72)	-0.022* (-2.30)	-0.024** (-2.63)		
Firm's CBG	-0.024 (-0.62)	-0.018 (-0.50)	0.007 (0.31)	0.007 (0.33)	0.007 (0.34)	0.006 (0.29)	0.006 (0.28)		
performance pressure	0.024*** (3.50)	0.033*** (8.04)		-0.030*** (-11.80)	-0.030*** (-11.84)	-0.031*** (-11.96)	-0.030*** (-11.78)		
Capital intensity					-0.002*** (-3.96)				
Corporate ownership	0.072* (1.98)					0.004 (0.98)			
CEO duality		0.003(0.49)			0.011** (2.96)		-0.003 (-0.73)		
Capital intensity * performance pressure									
Corporate ownership * performance pressure	0.140* (2.11)	-0.071 (-0.95)				0.066 (1.64)	0.012*** (0.26)		
CEO duality * performance pressure							0.021		
R^2	0.023	0.020	0.011	0.019	0.030	0.024	0.018		
Adjusted R^2	0.021	0.018	0.010	0.018	0.020	0.022	0.018		
F value	20.10***	20.02***	11.74***	12.85***	12.84***	12.73***	12.82***		

Table 7.

performance pressure and cross-international regional growth, but the moderating effect of corporate ownership is not significant. The above results are generally consistent with those of the related studies with a lag of one year, which indicates that the conclusions of this study have strong robustness.

5. Conclusion and discussions

In the process of making decision for firm's CBG, performance pressure is an important antecedent factor, and the state of managerial discretion also plays a differentially moderating role. The paper theoretically and empirically examines the influencing mechanism of performance pressure on the CBG of firms, as well as explores the moderating role of managerial discretion including three dimensions of capital intensity, ownership and CEO duality. The empirical results show that performance pressure has a significant positive effect on cross-industry growth and cross-domestic regional growth but has a significant negative effect on cross-international regional growth. Furthermore, managerial discretion also plays a differentially moderating effects on the relationship between performance pressure and CBG of firms. First, capital intensity strengthens the effect of performance pressure on cross-industry growth of firms. The reason mainly lies in the fact that firms with low capital intensity will invest less in fixed assets, investment more in products, have higher flexibility and are more likely to get involved in other industries in the face of performance pressure. Moreover, firms with low capital intensity have greater managerial discretion and are more inclined to implement cross-industry growth when faced with performance pressure, which not only can share resources and equipment but also expand sales channels and suppliers. Secondly, corporate ownership leads to a differentially moderating effect. Compared with state-owned enterprises, non-state-owned enterprises are more inclined to cross-domestic regional growth and are less likely to carry out cross-international regional growth when faced with performance pressure. This is mainly because non-state-owned enterprises are less subject to government intervention and political goals and can use their own advantages in business products for cross-domestic regional growth. Since the managers of non-state-owned enterprises are more sensitive to performance pressure, they will often take the initiative to choose more conducive to their own social and emotional wealth development and thus tend to choose cross-domestic regional growth so as to expand more market segments. Finally, CEO duality encourages firms not to implement cross-regional growth when faced with performance pressure but has no significant moderating effect on cross-industry and cross-domestic regional growth. This is mainly due to the fact that CEOs with dual roles usually have higher individual discretion within the firm, and other board members are more likely to be constrained in their strategic decisions, and thus the effectiveness of their strategies is more uncertain.

5.1 Implications for theory

First, by investigating how performance pressure affects firms' CBG, our research enriches relevant literature on understanding the driving factors of corporate growth at the organizational level. Due to the double-edged sword effect of performance pressure, the existing literature has drawn different research conclusions –some studies have shown that it enhances functional behavior (Eisenberger and Aselage, 2009; Zhang *et al.*, 2017b; Mitchell *et al.*, 2019), while others have shown that it produces dysfunctional behavior (Ellis, 2006; Gardner, 2012; Jensen *et al.*, 2019; Mai *et al.*, 2022). We grounded our theoretical explanation by bridging prospect theory, upper echelon theory and the literature on firm's growth strategies. Prospect theory provides a valuable perspective for explaining how performance pressures affect firms' decision-making. Despite extant research has explored factors

affecting firms' CBG, these studies mainly focus on organizational characteristics such as size, resources and networks (Ahammad *et al.*, 2016; Hitt *et al.*, 2006; Hoskisson and Hitt, 1990) and macro environment such as institutional environment (Boateng *et al.*, 2008; Du and Boateng, 2015). Because firms' CBG involves a high level of uncertainty and risk, requiring discretionary executive action and risk-taking tendency from decision-makers (Freeman and Cavusgil, 2007). By integrating and moving beyond this literature, our study illustrates how performance pressure affects firm's growth strategy.

Second, we propose three types of CBG and explore how they are affected by performance pressure. Our research highlights that despite high risk and unpredictability involved in CBG (Zaheer, 1995; Zhou and Guillen, 2016), firms under performance pressure are more likely to take this approach to growth. Although design and production are the major activities for manufacturing firms, we find that companies facing performance pressure tend to promote cross-industry growth and cross-domestic regional growth. Thus, our study depicts a richer picture of how performance pressure affects firms' growth strategies both internationally and domestically.

Third, we provide a more nuanced and accurate model summarizing how the relationship between performance pressure and firms' CBG depends on managerial discretion, thus revealing the conditions under which this may occur. Previous studies have focused less on the interaction between managerial discretion and performance pressure (Wangrow *et al.*, 2015; Baixauli-Soler *et al.*, 2021), and we have contributed to the strategic literature on this topic. This paper constructs a theoretical model by dividing the dimensions of managerial discretion so that the empirical results can be better verified. This can also be well explained from the perspective of entrepreneurship: firms' CBG is equivalent to the second venture of enterprises (Shane and Venkataraman, 2000; Klueter and Monteiro, 2017), and the availability of resources such as capital, contacts and policy support will affect the choice of cross-boundary strategies. An in-depth understanding of managerial discretion is the key to clarifying the relationship between performance pressure of enterprises in emerging markets and corporate strategy, which is helpful to understand the overall influence mechanism of performance pressure and can provide theoretical framework and inspiration for subsequent research on firms' CBG.

5.2 Implications for practice

First, we provide CBG strategies that companies can follow. In the VUCA era, there is a trend of accelerated convergence between the traditional industries and emerging industries, and competition among firms in the same industry or geographic region becomes more intense, which makes firms take the initiative or be forced to find new growth space. CBG has become a powerful way for enterprises to expand market segments, build core competitiveness and improve performance. This paper puts forward a variety of CBG strategies according to the cross-boundary scope of enterprises, as well as the senior executives should timely adjust the strategic focus and thinking of business development. Second, top executives should dialectically consider the double-edged effect of performance pressure on firm's CBG. We generally believe that when a company faces performance pressure, it will reduce the highly uncertain strategic behavior of "cross-boundary." However, our research results show that, when firms are under pressure for performance, they will still choose CBG to achieve short-term asset scale expansion and investment risk diversification. Therefore, firms cannot simply copy and graft across different industries or regions and must comprehensively consider the double-edged sword effect of performance pressure to judge what kind of cross-boundary strategy needs to be implemented. Finally, firms should create an appropriate environment of managerial discretion. Since managerial

discretion significantly moderates the relationship between performance pressure and firm's CBG, it is essential for senior managers to make and implement the strategic decisions. Thus, firms should also create a suitable internal governance environment so as to play the positive role of managerial discretion and improve the quality of decision-making.

5.3 Limitations and future research directions

First, this study examines performance pressure from the level of the firm but not enough from the level of employees. [Mitchell et al. \(2019\)](#) pointed out that an organization's success is only as good as its employees' productiveness; this reality contributes to an urgency (pressure) for employees to perform well. Performance pressure can serve as a source of stress for employees. This view assumes that performance pressure not only comes from the influence of the organization but may also be perceived by employees to meet their own and organizational needs under the joint influence. Thus, future research can integrate the influence of performance pressure at the organizational and employee levels, and design the measurement indicators to explore the interaction effect of multiple performance pressure on the CBG within one firm. Second, our data set relies on panel data of publicly listed firms between 2013 and 2016. Future research should validate our findings when new and systematic data becomes available. Finally, this study has shown that looking at performance pressure has the potential to enrich our understanding of how performance factors interpret firm's strategic actions. However, firms can have multiple missions or concerns when implementing the decision of CBG, including for being diversified or focused, environmentally friendly or high-technology oriented. Those multiple concerns might lead to fuzziness in the minds of investors and customers as to what might be expected of the firm ([Parker et al., 2019](#)). Therefore, a fruitful next step would be to study how the extent and direction of firm's CBG are shaped by alliance activities outside firm boundaries when they evaluate the strategic initiatives.

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